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Integrating RUSLE, AHP, GIS, and cloud-based geospatial analysis for soil erosion assessment under mediterranean conditions

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Soil erosion is a major environmental challenge in Mediterranean regions, where climatic variability, steep slopes, and human activities accelerate land degradation. In the north-central region of Algeria, the Mitidja Plain faces increasing erosion pressure, threatening biodiversity, agricultural productivity, and long-term soil sustainability. This study aims to assess soil erosion risk by integrating the Revised Universal Soil Loss Equation (RUSLE), the Analytical Hierarchy Process (AHP), and Geographic Information System (GIS) techniques within a Cloud-Based Geospatial (CBG) framework using the Google Earth Engine (GEE) platform. High-resolution datasets on rainfall, topography, soil properties, and land cover were processed in GEE to derive five RUSLE factors: rainfall runoff erosivity (R^E), soil erodibility (K^S), slope length steepness (L^S), cropping management (C^M), and management practices (P^C). The analysis revealed that 41% of the Mitidja Plain is at severe erosion risk, with an average soil loss of 88.72 t ha^{-1} yr^{-1} and a maximum of 161.13 t ha^{-1} yr^{-1}. Erosion hotspots correspond to areas where slopes exceed 22°, vegetation cover is sparse, and rainfall intensity is high. The AHP-weighted integration achieved strong predictive accuracy (AUC = 0.87), identifying slope characteristics as the most influential factor (weight = 0.292). Forested areas reduced erosion risk in 30% of the region, while unprotected mountainous zones covering 22% of the study area require urgent intervention. These findings demonstrate the effectiveness of CBG-enhanced modeling for mapping priority conservation areas. Recommendations include terracing, check dams, vegetation restoration, and adaptive agricultural practices to reduce soil loss, particularly in agricultural lands with moderate to high vulnerability (48% of the plain). The methodology provides a replicable framework for other Mediterranean regions facing similar erosion pressures, offering robust spatial data to guide soil management and conservation planning.

Keywords Land degradation, Soil erosion, RUSLE, GEE; AHP, Mitidja plain, Algeria

Soil erosion is a major driver of land degradation that severely impacts soil fertility, agricultural productivity, and ecosystem health[1]. It leads to the loss of nutrient-rich topsoil essential for crop growth and contributes to sedimentation in water bodies, resulting in degraded water quality and increased flood risk[2,3]. Human activities such as intensive agriculture, deforestation, and mining have accelerated soil erosion, further threatening food security and economic stability, particularly in regions dependent on agriculture[4–7].

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As a complex environmental process, soil erosion not only reduces soil organic matter and disrupts soil structure but also exacerbates nutrient depletion and drainage issues, thereby diminishing land productivity and ecosystem resilience[8–10]. Efficient soil conservation requires a thorough understanding of erosion dynamics and the identification of vulnerable areas through reliable modeling approaches[11]. Soil erosion models play a crucial role in supporting decision-makers by simulating erosion patterns, predicting soil loss rates, and assessing conservation strategies to mitigate land degradation[12].

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Among existing models, the Revised Universal Soil Loss Equation (RUSLE) offers improved accuracy over the original USLE for predicting soil erosion at broader spatial scales[15,16]. It integrates key factors including rainfall erosivity (R^E), soil erodibility (K^S), slope length and steepness (L^S), cropping management (C^M), and management practices (P^C) to estimate soil loss[17,18]. The application of Geographic Information Systems (GIS) and remote sensing enables efficient data integration and spatial analysis to support RUSLE, providing detailed erosion risk mapping over large areas[19–22].

The Analytical Hierarchy Process (AHP), when combined with GIS, provides a systematic multicriteria evaluation framework that weights and prioritizes the factors influencing erosion risk. Previous studies have demonstrated the effectiveness of integrating RUSLE and AHP within GIS for soil erosion assessment in Mediterranean and other semi-arid regions[23–26]. However, few studies have leveraged cloud-based platforms such as Google Earth Engine (GEE) to enhance data processing efficiency and scalability for regional erosion modeling[27,28].

GEE offers access to extensive satellite datasets and powerful computational resources, facilitating large-scale soil erosion assessments by implementing models like RUSLE in a cloud environment[29–31]. While research in other countries has integrated RUSLE, AHP, and GIS for erosion mapping, there remains a gap in applying this combined approach using GEE specifically to the Mitidja Plain in Algeria a Mediterranean region facing significant erosion challenges due to its climatic and land use conditions[32].

While this integrated approach is powerful, it is important to acknowledge its limitations. The accuracy of the model is contingent on the quality and resolution of the input data, and the empirical nature of RUSLE may not capture all non-linear erosion processes. Furthermore, the proposed management strategies require validation through long-term field studies to assess their real-world applicability and sustainability[33,34].

This study aims to fill this gap by integrating RUSLE, AHP, and GIS within the GEE platform to assess soil erosion risk in the Mitidja Plain under Mediterranean conditions. The objectives are to identify erosion hotspots, evaluate the relative importance of controlling factors, and recommend targeted conservation measures to support sustainable land management tailored to the region's unique environmental context.

Materials and methods

Description of the study area

The Mitidja Plain, located in north-central Algeria between 36°30' and 36°45'N and 2°30' and 3°00'E, covers an area of about 2561 hectares (ha)[35] and is bordered by the Mediterranean Sea to the north and the Tell Atlas Mountains to the south, as shown in Fig. 1. It is characterized by fertile Quaternary alluvial deposits[36,37], a Mediterranean climate with 600–800 mm annual rainfall[38], and extensive agricultural activity including cereals, fruits, and vegetables. Despite its importance as one of Algeria's most productive agricultural regions, the plain faces severe soil erosion risks due to steep slopes, intensive land use, and rainfall variability[39].

The study methodology

The GEE-based approach delivered a comprehensive spatial analysis of large-scale geospatial data for soil erosion risk across the Mitidja plain, combined with the AHP technique that was applied to analyze different environmental parameters and involves ranking based on certain criteria to solve a very complex decision-making process were used to estimate average soil loss (A) using the RUSLE equation. This equation is calculated using specific factors for the study area, including rainfall-runoff erosivity (R^E), soil erodibility (K^S), slope length steepness (L^S), cropping management (C^M) and management practices (P^C), these factors represent the specific environmental and land-use conditions that contribute to the overall soil erosion hazard assessment over the investigated area. This study methodology was outlined in the following flowchart (Fig. 2).

The data Preparation and processing

In this study, data from the Sentinel-2, Multispectral Instrument (MSI) Level-1 C (COPERNICUS/S2_SR), provided by the European Space Agency (ESA, http://www.esa.int/Our_Activities/Observing_the_Earth/Copernicus/Overview4), and the MODIS sensor onboard NASA's Terra and Aqua satellites (http://search.earthdata.nasa.gov/search) were utilized, accessed on 16 May 2024. Sentinel-2 offers wide-swath, high-resolution (10 m) multispectral images that support land monitoring for applications such as vegetation, land use/land cover assessment, and the observation of inland and coastal water bodies[40]. MODIS data, known for its wide spectral range and cost-effectiveness, is frequently used for monitoring vegetation health and surface conditions and it is particularly useful for studying soil erosion due to its timely availability and ability to provide frequent, medium-resolution data[41]. MODIS images undergo several corrections such as radiometric, geometric, and atmospheric to ensure their accuracy. In this study, MODIS/061/MCD12Q1 data with an 8-day time resolution and 500 m spatial resolution were employed to calculate soil erosion; cropping management and supporting conservation practices factors (30 m) using the GEE platform based on AHP technique.

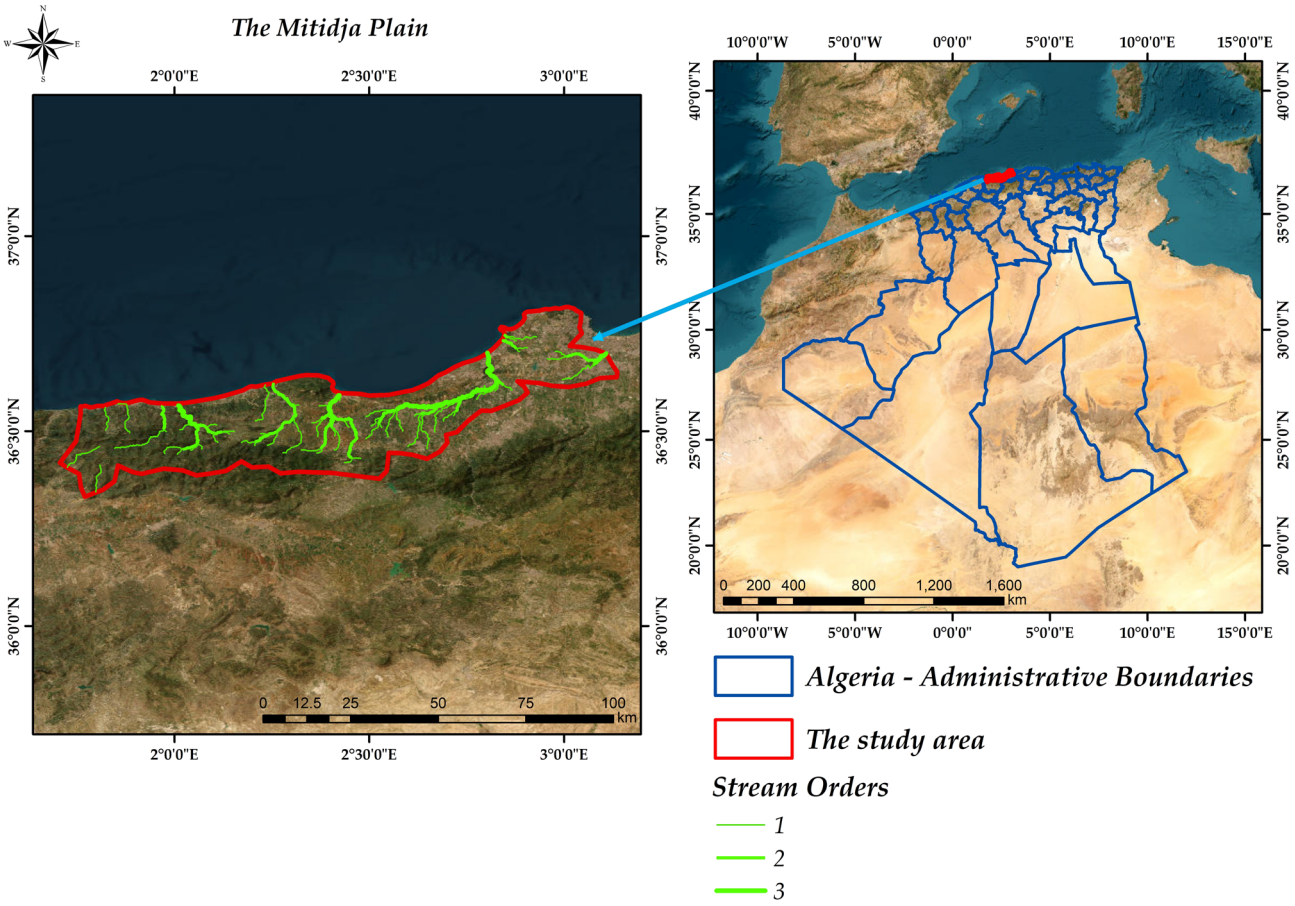

Fig. 1. Location map of the study area (Generated by ArcGIS 10.2.2. software; Esri (Environmental Systems Research Institute, Redlands, CA, USA; https://support.esri.com/en-us/patches-updates/2016/arcgis-10-2-2-for-desktop-engine-server-update-geopacka-2093).

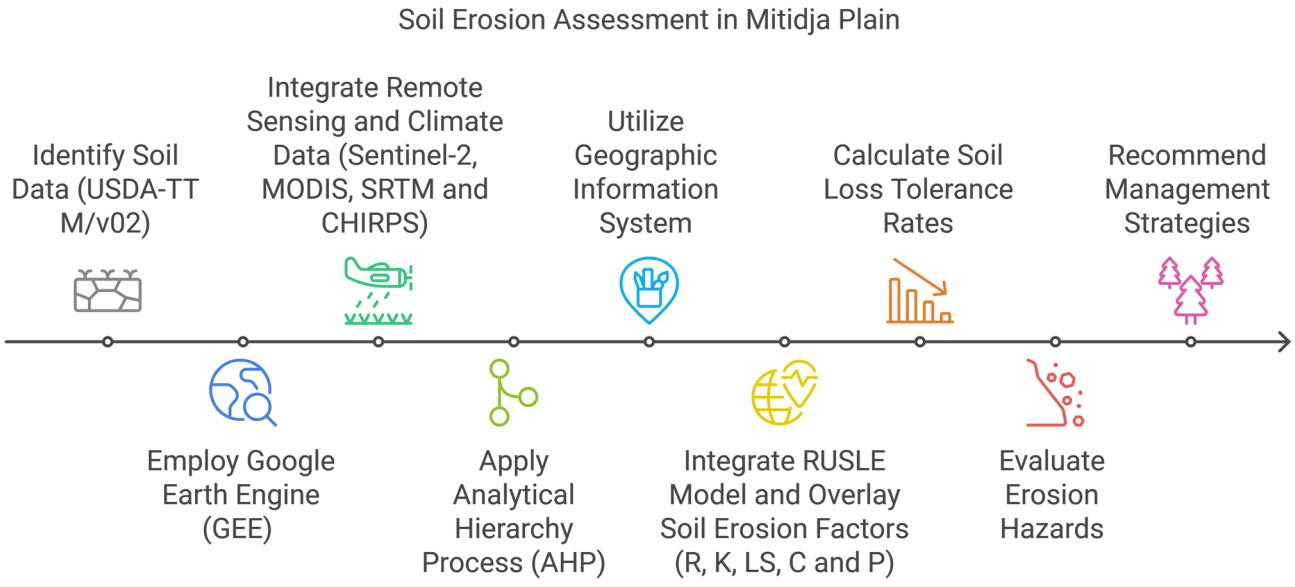

Fig. 2. The study methodological framework.

Factor	Management strategies
R^E	Unchangeable factor for a field
K^S	Unchangeable factor for a field
L^S	Terraces may be reducing the slope length Terracing requires additional investment
C^M	Minimizing C factor will lead to reduced soil erosion Using cropping and tillage systems where possible
C^P	Lowest P factor will result in lower soil losses Using support practices such as cross-slope farming

Table 1. Soil erosion factors and management strategies to reduce soil losses.

Soil erosion class	Soil loss rate (t ha^{-1} yr^{-1})
Severe	> 100
Very High	70–100
High	40–70
Moderate	10–40
Slight	< 10

Table 2. Soil erosion classes and soil loss tolerance rates.

The key variables used in this soil erosion study include: (a) annual rainfall data obtained from the Climate Hazard Group Infrared Precipitation with Station Data (CHIRPS) (0.05°, approximately 5 km) database (https://chc.ucsb.edu/data/chirps), which was used to calculate the rainfall runoff erosivity factor (30 m)[42], (b) slope length and steepness factors derived from the Shuttle Radar Topography Mission (SRTM, USGS/SRTMGL1_003) digital elevation data with a resolution of 1 arc-second (approximately 30 m), available on a near-global scale (https://earthexplorer.usgs.gov/), (c) land use and normalized difference vegetation index (NDVI) determined from Sentinel-2 imagery, and (d) soil type and characteristics collected from (USDA-TTM/v02) soil texture classes at six different soil depths (0, 10, 30, 60, 100, and 200 cm) with a resolution of 250 m (https://websoilsurvey.nrcs.usda.gov/app/).

Additionally, ArcGIS 10.2.2 software was used to create maps illustrating location map of the study area, soil erosion factors thematic maps, annual soil loss (RUSLE model) map and soil erosion potential hazards map.

Soil erosion thematic maps computation: RUSLE model

To estimate, predict, quantify, and assess the annual average soil loss rate the average annual soil loss per unit area (A, t ha^{-1} yr^{-1}) for a given area under particular management conditions, and estimate the soil erosion process spatial mapping using GEE-based approach, the RUSLE was utilized as widely used soil erosion empirical-based model, relies on various factors such as rainfall-runoff erosivity (R^E MJ mm ha^{-1} h^{-1} yr^{-1}), soil erodibility (K^S t ha MJ^{-1} mm^{-1}), slope-length (L) and slope-steepness (S) factors (L^S), cover and management (C^M), and conservation practices (C^P), as numerical representation of the particular conditions that influence the soil erosion risk in the region under study, and the L^S, C^M, and C^P factors are dimension less values.

According to Renard, 1997[43], the RUSLE model was used in the current study for soil erosion assessment through the following Eq. 1:

$$A = R^E \times K^S \times L^S \times C^M \times C^P \tag{1}$$

Based on McKague, 2023[44], soil erosion values and tolerance rates are closely linked to both soil characteristics and management practices, as shown in Tables (1 and 2).

Rainfall-runoff erosivity factor (R^E)

Based on GEE-based approach, the rainfall-runoff erosivity R^E factor plays a crucial role in influencing potential sheet and rill soil erosion and quantifies soil erosivity based on the aggressiveness of daily basis runoff sediment deposits in the study area over a given time period (2001–2023)[45], and it was considered as a major challenging task due to unavailability of high-resolution rainfall-runoff database for the investigated area. The R^E factor was calculated using the linear relationship empirical Eq. 2, that was developed by Choudhury and Nayak, 2003[46].

$$R^E = 79 + (0.363 * X_a)) + 89.7 \tag{2}$$

where R^E is the rainfall-runoff erosivity factor and X_a is the annual precipitation average (mm).

The soil erodibility factor (K^S)

The K^S factor is one of the most significant barriers for soil erosion modeling on large spatial scales is insufficient data on soil attributes[47]. K^S factor describes the predisposition of soil particles to be transported and separated

by runoff and plays a crucial role in evaluating the inherent vulnerability of soil to erosion caused by rainwater and runoff[48].

The K^S factor is rated primarily on a scale of 0 to 1, with higher values indicating a greater susceptibility to soil erosion, relies on various soil characteristics encompassing mineralogical, physical, chemical, and morphological features such as soil texture classes, soil structure, organic matter content, soil permeability, and infiltration.

In this study of the Mitidja plain, we employed CBG analysis using GEE platform to estimate the K^S factor, utilized soil types and texture maps to conduct geospatial analyses efficiently, providing a robust and scalable approach. This approach considers local variations and characteristics, accurately representing soil erosion factors within the study area.

The K^S factor was calculated by means of the following Eqs. 3 and 4, which were developed from global data of measured K^S values by Renard, 1997[43]:

$$K^S = 0.0034 + 0.0405\,\mathrm{xexp}\left[-0.5\left(\frac{logD_g + 1.659}{0.7101}\right)^2\right] \tag{3}$$

$$D_g = \exp\left(\sum f_i\, In\, \left(\frac{d_i + d_{i-1}}{2}\right)\right) \tag{4}$$

where D_g is the particle size geometric mean, for clay, silt, sand texture classes, d_i is particle size maximum diameter (mm), d_{i-1} is particle size minimum diameter, and f_i is the mass fractions.

McKague, 2023[44] clarified the K^S factor rates for various soil textures corresponding to organic matter content, as represented in Table 3.

Topographic (slope-length (L) and slope steepness (S) factors (L^S))

The L^S factor in RUSLE is a crucial component in assessing soil erosion rates and expressed as the ratio extracted from the SRTM satellite data, integrates slope length (L) and slope steepness (S) into a unified index by considering the distance from the runoff initiation point to the deposition area using GEE, CBG analysis computation[49]. L^S factor reflects the influence of topography on soil erosion, standardized at a length of 22.1 m and a steepness of 9%, demonstrated that increases in slope length and slope steepness with the greatest accumulation can produce higher overland flow velocities, increases transport detachment and capacity with correspondingly higher erosion[50].

The L^S factor was defined by Shinde et al., 2011[51] and expressed as the following Eq. 5.

$$L^S = \left(\frac{L}{22.1}\right) * \left(65.4*(sin\theta)^{\,2} + 4.56*sin\theta + 0.0065\right) \tag{5}$$

where L is the slope length (m), and S is the slope gradient (%).

The L^S factor rates were clarified for various topographic factors by McKague, 2023[44], shown in Table 4.

Textural Class	K^S factor		
	Average OMC	Less than 2% OMC	More than 2% OMC
Clay	0.22	0.24	0.21
Clay loam	0.30	0.33	0.28
Coarse sandy loam	0.07	–	0.07
Fine sand	0.08	0.09	0.06
Fine sandy loam	0.18	0.22	0.17
Heavy clay	0.17	0.19	0.15
Loam	0.30	0.34	0.26
Loamy fine sand	0.11	0.15	0.09
Loamy sand	0.04	0.05	0.04
Loamy very fine sand	0.39	0.44	0.25
Sand	0.02	0.03	0.01
Sandy clay loam	0.20	–	0.20
Sandy loam	0.13	0.14	0.12
Silt loam	0.38	0.41	0.37
Silty clay	0.26	0.27	0.26
Silty clay loam	0.32	0.35	0.30
Very fine sand	0.43	0.46	0.37
Very fine sandy loam	0.35	0.41	0.33

Table 3. Soil texture and corresponding soil erodibility factor rates. *OMC; organic matter content.

Slope length (m)	Slope (%)	L^S factor	Slope length (m)	Slope (%)	L^S factor
15	10	0.98	61	10	1.95
	8	0.70		8	1.41
	6	0.48		6	0.95
	5	0.38		5	0.76
	4	0.30		4	0.53
	3	0.22		3	0.39
	2	0.16		2	0.25
	1	0.10		1	0.16
	0	0.06		0	0.08
30.5	10	1.38	122	10	2.76
	8	1.00		8	1.99
	6	0.67		6	1.35
	5	0.54		5	1.07
	4	0.40		4	0.70
	3	0.30		3	0.52
	2	0.20		2	0.30
	1	0.13		1	0.20
	0	0.07		0	0.09
46	10	1.70	244	10	3.90
	8	1.22		8	2.82
	6	0.83		6	1.91
	5	0.66		5	1.52
	4	0.47		4	0.92
	3	0.35		3	0.68
	2	0.23		2	0.37
	1	0.15		1	0.24
	0	0.08		0	0.11

Table 4. Slope length and slope steepness data and corresponding L^S factor rates.

Crop cover and management factor (C^M)

The Crop cover and management factor C^M is used to determine the impacts of cropping and management practices in agricultural management and is notably susceptible to human influence to mitigate soil erosion[52]. Sentinel-2 data from Google Earth Engine (GEE) were employed to address land cover data based on the maximum likelihood algorithm and ground truth verification points. This approach leverages the geospatial cloud-based analysis capabilities, offering a more efficient method for land cover and management in the Mitidja plain that is very sensitive to potential tilled soil erosion under Mediterranean conditions to the consequent loss from continuous fallow land. the C^M factor together with L^S factor is most sensitive to soil loss. In this study, the C^M factor was evaluated through the most widely used satellite derived normalized difference vegetation index (NDVI) indicator, as an indicator of the energy reflected by the earth related to various cover type conditions following the empirical Eqs. 6 and 7[53].

$$C^M = \exp[\alpha \frac{NDVI}{(\beta - NDVI)}] \tag{6}$$

$$NDVI = \frac{NIR - R}{NIR + R} \tag{7}$$

where C^M is the crop cover and management factor; α and β are the parameters that determine the shape of the curve relating to NDVI; NIR is near infrared band and R is red band.

The C^M factor rates for crop management values of the present study ranging from 0 to 1, where a lower C^M suggests minimal soil loss that was assigned to a few pixels with negative values, and the maximum value indicates higher susceptibility to soil loss that was assigned to pixels with value greater than 1, as explained by McKague, 2023[44] and shown in Tables 5 and 6.

Conservation practice factor (C^P)

The C^P factor represents the effectiveness of management practices and soil loss ratio of up-and down-slope tillage through vegetation caver, soil biomass, and runoff control, as determined by agronomical practices[54]. The C^P factor was calculated using remote sensing-driven analyses methodology in GEE-based approach based on Sentinel-2 image by defining a function that extracts relevant bands or properties for conservation practice, allowing for a comprehensive assessment of soil erosion susceptibility[43].

Crop Type	C^M factor
Grain corn	0.40
Silage corn, beans and canola	0.50
Cereals (spring and winter)	0.35
Seasonal horticultural crops	0.50
Fruit trees	0.10
Hay and pasture	0.02

Table 5. Crop cover and management factor (C^M) rates.

Tillage Method	C^M factor
Fall plow	1.0
Spring plow	0.90
Mulch tillage	0.60
Ridge tillage	0.35
Zone tillage	0.25
No-till	0.25

Table 6. Tillage method (C^M factor) rates.

Support Practice	C^P Factor
Up and down slope	1.0
Cross slope	0.75
Contour farming	0.50
Strip cropping, cross slope	0.37
Strip cropping, contour	0.25

Table 7. Conservation practice factor C^P rates.

The C^P factor rates for conservation support practices, as represented by McKague, 2023[44], are shown in Table 7.

Multi-criteria decision using analytical hierarchy process (AHP)

AHP technique is a multi-criteria decision analysis method which applied in this study by integrating multiple factors and involves ranking based on certain criteria to solve a very complex decision-making problems, based on its interdependency and interrelationship between other parameter, and hierarchy model was done through higher to the lower level using a pairwise comparison of a different parameter which influences the soil erosion process expressed in a numerical scale and consistency ratio[30,55], as represented in the following Eqs. 8 and 9:

$$M = \begin{bmatrix} 1 & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & 1 & a_{22} & \ldots & a_{2n} \\ a_{31} & a_{32} & 1 & \ldots & \cdots \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \ldots & \ldots & 1 \end{bmatrix} \tag{8}$$

where

$$a_{ij} = \frac{Weight\ of\ attribute\ i}{Weight\ of\ attribute\ j} \tag{9}$$

where a is the weight may vary in the numerical scale of 1 and 9 and the element of 1/1 in the matrix have equal importance for both parameters i and j based on a preference scale (SN) that represented in Table 8.

The present study considered six individual parameters such as soil texture, OMC, slope, crop cover, tillage method, and support practice of the study area that have significant influence on the soil erosion. The highest eigenvalue of the matrix λ_{max} for each factor was determined and computed by adding each column values of pairwise comparison matrix[30].

SN	1	2	3	4	5	6	7
Intensity of importance	1	3	5	7	9	2,4,6 and 8	Reciprocals
Definition	Equal importance	Weak importance	Strong importance	Demonstrated importance	Absolute importance	Intermediate values	Compared activities
Explanation	Activities are equally to the objectives	Slightly activity	Strongly activity	Activity is strongly in practice	Activity is highest possible	Compromise is needed	

Table 8. Scale of preference for the analytical hierarchy process (AHP) pairwise comparison by Saaty, 1989[29].

The consistency of the pairwise comparison matrix of order n was calculated through the consistency index (CI) and consistency ratio (CR) that should be less than or equal to 0.1 to be accepted as consistent[56], as represented in the following Eqs. 10 and 11:

$$CI = \frac{\lambda_{max} - n}{n - 1} \tag{10}$$

$$CR = \frac{CI}{RI} \tag{11}$$

where RI is the average of resultant consistency index of randomly generated comparison and it varies according to order of the matrix[30]. The CR >0.1 endorses the omission of variable and CR <0.1 will allow the variables in the analysis.

When conducting comparisons, a compatibility ratio of 0.1 or lower signifies compatibility. Following this, a map depicting soil erosion was generated using CBG analysis in GEE platform based on the next specified Eq. 12:

$$A = (R^E \times \mathrm{W}_i) \times (K^S \times \mathrm{W}_i) \times (L^S \times \mathrm{W}_i) \times (C^M \times \mathrm{W}_i) \times (C^P \times W_i) \tag{12}$$

where W_i is the weight of each soil erosion factor.

Results

Rainfall-runoff erosivity factor (R^E)

The CHIRPS rainfall data analysis indicated that the study area's average annual rainfall ranges from 830 to 2250 mm. The spatial distribution of R^E factor showed significant variability across the region, with values ranging from 583.86 to 1072.26 MJ mm ha^{-1} h^{-1} yr^{-1} and an average of 828.06 MJ mm ha^{-1} h^{-1} yr^{-1} (Fig. 3). The western part of the study area exhibited higher R^E values, which gradually decreased toward the southeast. This pattern is likely attributed to the southeastern leeward side of the mountain, which acts as a barrier to the southwest monsoonal rainfall, potentially causing higher rainfall in other parts of the study area.

The soil erodibility factor (K^S)

The K^S factor, which reflects the vulnerability of soil particles to detachment and transport by runoff, is influenced by various soil properties such as texture and organic matter content. Using the RUSLE model, GIS, and satellite imagery based on cloud system, the study assessed the K^S factor across the Mitidja plain, revealing spatial variability due to heterogeneous soil characteristics. The resulting K^S factor map identified erosion-prone zones requiring prioritized conservation efforts, with values ranging from 0 to 0.0394 and an average of 0.0197 t ha MJ^{-1} mm^{-1}, reflecting diverse soil textures such as loamy and sandy clay loam (Fig. 4).

Topographic factors (L^S)

The L^S factor, a key element in soil erosion evaluation, accounts for the combined impact of slope length and steepness on erosion rates. Calculated using Eq. (5), this factor exhibits significant spatial variation across the Mitidja plain, reflecting the area's diverse terrain. Higher L^S values correspond to steeper slopes and longer flow paths, marking regions with increased erosion risk. The L^S factor map (Fig. 5) helps identify critical zones requiring targeted erosion control measures. The wide range of values from 1.7 to 4110 highlights the need for customized soil management strategies based on local topography. Analysis using SRTM-DEM slope data showed that gentler slopes (0–22°) had lower L^S values (0–0.983.983), while steeper areas exhibited much higher values (0.872–1.411), underscoring the importance of slope-specific conservation efforts to ensure sustainable land use in the Mitidja plain.

Cover management factor (C^M)

The C^M factor, a crucial component in soil erosion assessment, reflects the influence of land cover and agricultural practices on soil protection. Using Sentinel-2 derived NDVI data, this study quantified the C^M factor, revealing its spatial variability across the Mitidja plain. NDVI served as a key indicator of vegetation density, with calculated C^M values ranging from < 0.155 (indicating dense vegetation cover) to 0.915 (representing bare or rocky terrain). The results demonstrate that areas with lower C^M values correspond to well-vegetated zones that effectively reduce erosion risk, while higher values mark exposed, vulnerable landscapes. These findings align with similar studies showing that vegetation plays a critical protective role against soil erosion. The spatial distribution of C^M

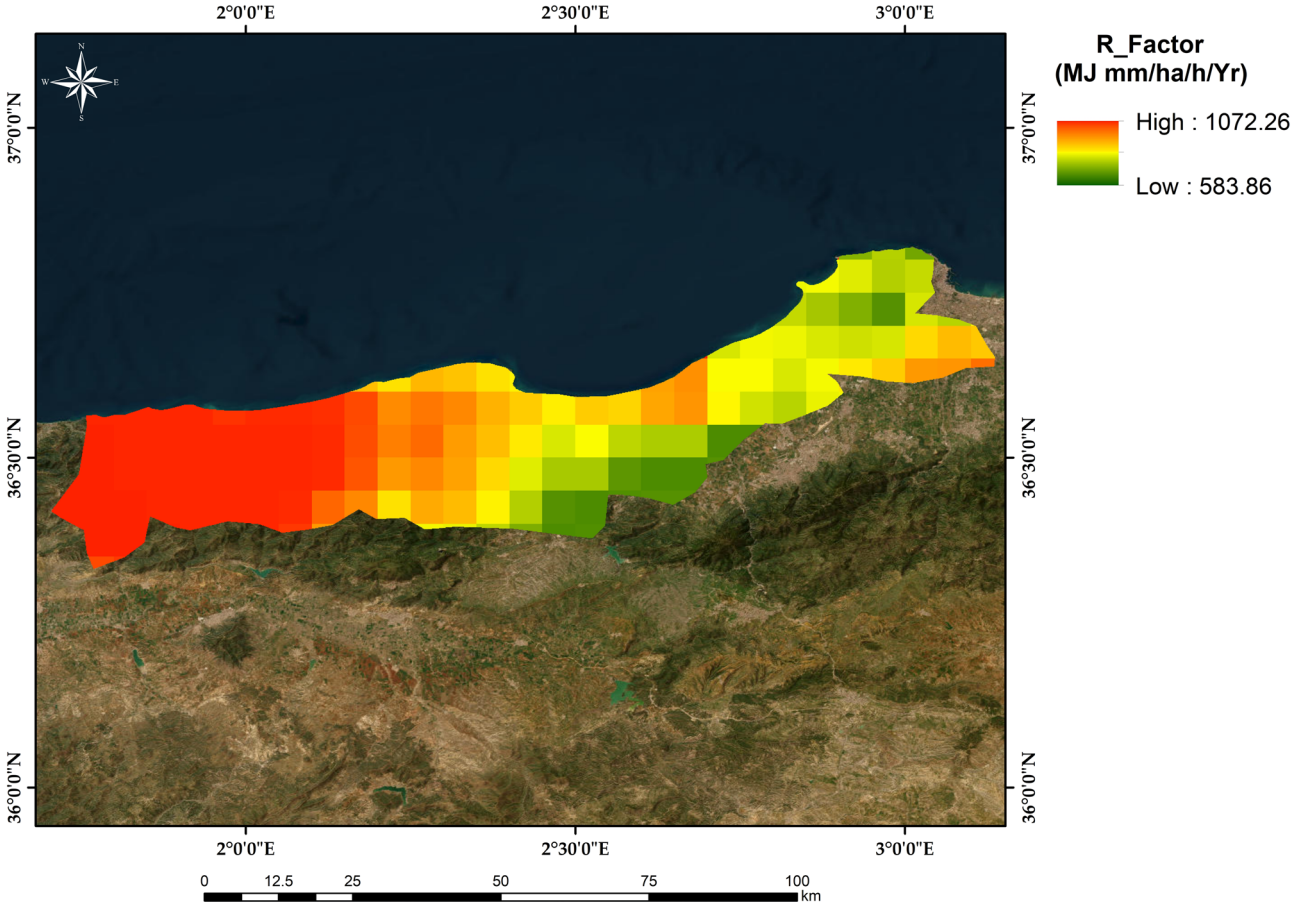

Fig. 3. The spatial distribution of rainfall-runoff erosivity (R^E) factor.

values provide essential insights for targeted land management, suggesting that erosion control measures should prioritize areas with minimal vegetative cover (Fig. 6).

Conservation practice factor (C^P)

The C^P factor, which ranges from 0 to 1, reflects the effectiveness of erosion control measures across the landscape (Fig. 7). Lower C^P values indicate well-managed areas with implemented conservation practices, such as settlements, while higher values ($C^P = 1$) signify regions lacking protective measures. In this study, the northern and western parts of the Mitidja plain exhibited the highest C^P values (1) due to rocky mountainous terrain, sparse snow cover, and barren lands where no conservation practices were applied. Similarly, dense forests, though naturally vegetated, were assigned a C^P value of 1 since they lack active human-led management interventions. Water bodies were excluded from the analysis as they inherently have a C^P value of 0. These findings highlight the uneven distribution of conservation efforts, emphasizing the need for targeted soil protection strategies, particularly in erosion-prone, high C^P zones.

Soil loss estimation: RUSLE model

The annual soil loss rate across the Mitidja plain was estimated by integrating all five RUSLE factors using cloud-based techniques, suggesting an average erosion rate of 88.72 t ha^{-1} yr^{-1}, with values ranging from 38.32 to 161.13 t ha^{-1} yr^{-1} (Table 9; Fig. 8). The analysis classified soil erosion into varying susceptibility levels: moderate (37% of the area, 942 ha), high (13%, 337 ha), very high (9%, 222 ha), and severe (41%, 1060 ha), as represent in Fig. 9. The spatial patterns indicate that moderate erosion zones require targeted conservation measures, while high and very high-risk areas demand more extensive mitigation strategies. The classification of 41% of the plain under severe erosion risk highlights areas where soil degradation is potentially critical and where conservation efforts should be prioritized. These findings emphasize the spatial variability of erosion risk and highlight the need for prioritized, site-specific conservation planning to address soil loss effectively.

Soil erosion potential hazard assessment

The soil erosion potential zones across the Mitidja plain were mapped by integrating the RUSLE model with the AHP technique in ArcGIS, evaluating five key erosion factors: rainfall-runoff erosivity, soil erodibility, slope characteristics, land cover management, and conservation practices. Through AHP's pairwise comparison matrix, each factor was statistically weighted, with normalized eigenvector values ranging from 0.061 to 0.292 (Table

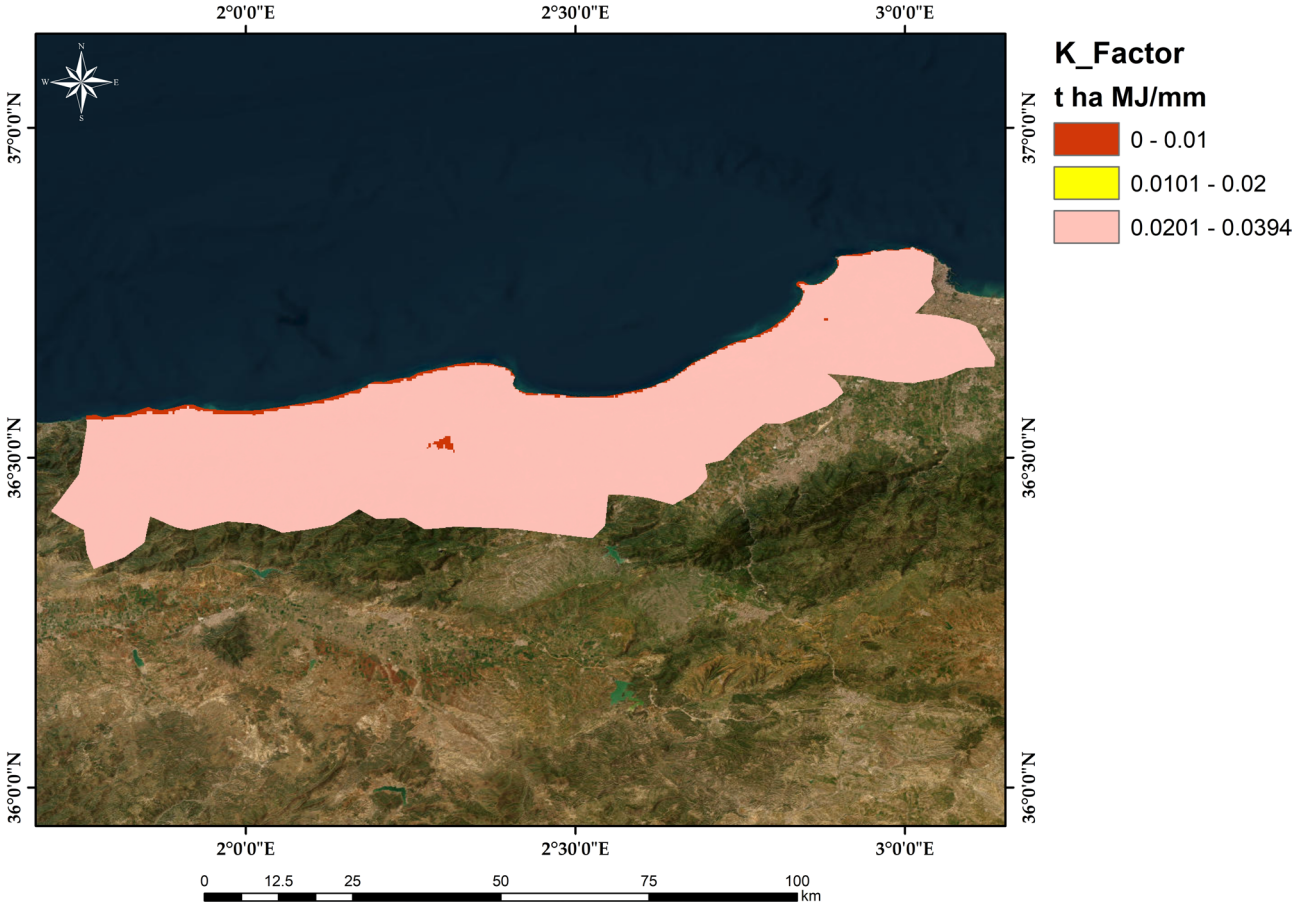

Fig. 4. The spatial distribution of the soil erodibility (K^S) factor.

10), revealing their relative influence on erosion vulnerability. The analysis demonstrated that topographic and rainfall-runoff erosivity factors carried the highest weights, indicating their dominant role in soil loss patterns. These multi-criteria approach effectively identified critical erosion hotspots where steep topography coincides with intense rainfall and poor vegetation cover. The resulting susceptibility map provides a robust foundation for prioritizing conservation measures, particularly in high-weight areas where multiple risk factors converge.

The soil erosion hazard map (Fig. 10) classified the Mitidja plain into four distinct vulnerability categories; low, moderate, high, and critical through an integrated assessment of weighted erosion parameters. Results revealed a clear spatial pattern, with critical zones (22%) of the study area, predominantly located in steep-sloped northern and western sectors where high rainfall erosivity, erodible soils, and minimal vegetation cover converge. Moderate to high vulnerability classes covered nearly 48% of the plain, typically occurring in agricultural lands with moderate slopes but inadequate conservation practices. The remaining 30% showed low susceptibility, corresponding to flat terrains with protective vegetation. These findings demonstrate the effectiveness of multi-parameter weighting systems in erosion risk assessment, with critical zones showing 3–5 times greater erosion potential than low-risk areas. The spatial patterns align with field observations of severe gullying in northern sectors, validating the model's accuracy.

The soil erosion hazard zonation map revealed varying degrees of vulnerability across the study area, ranging from low to critical, with forest cover playing a significant protective role in many areas (Table 11). However, zones classified as high to critical erosion risk were predominantly found in steeply sloped regions characterized by dense drainage networks and abundant lineaments, suggesting strong geomorphological control on erosion processes. The spatial analysis showed the critical hazard class covered the largest area (1812 ha), followed by high-risk zones (260 ha), while low-vulnerability areas were most limited (239 ha). The disproportionate distribution of critical zones underscores the need for targeted slope stabilization measures in high-risk areas, particularly where erosive forces are amplified by geological lineaments.

Area under curve (AUC)-based model validation

The validation of the AHP-based soil erosion assessment model demonstrated strong predictive accuracy, achieving AUC value of 86.96% in the receiver operating characteristic analysis (Fig. 11). This high accuracy score confirms the model's effectiveness in identifying erosion-prone areas when incorporating key physical parameters of the Mitidja plain. The results suggest that the selected factors including slope characteristics, land cover, and soil properties appropriately captured the complex interactions driving erosion processes in the

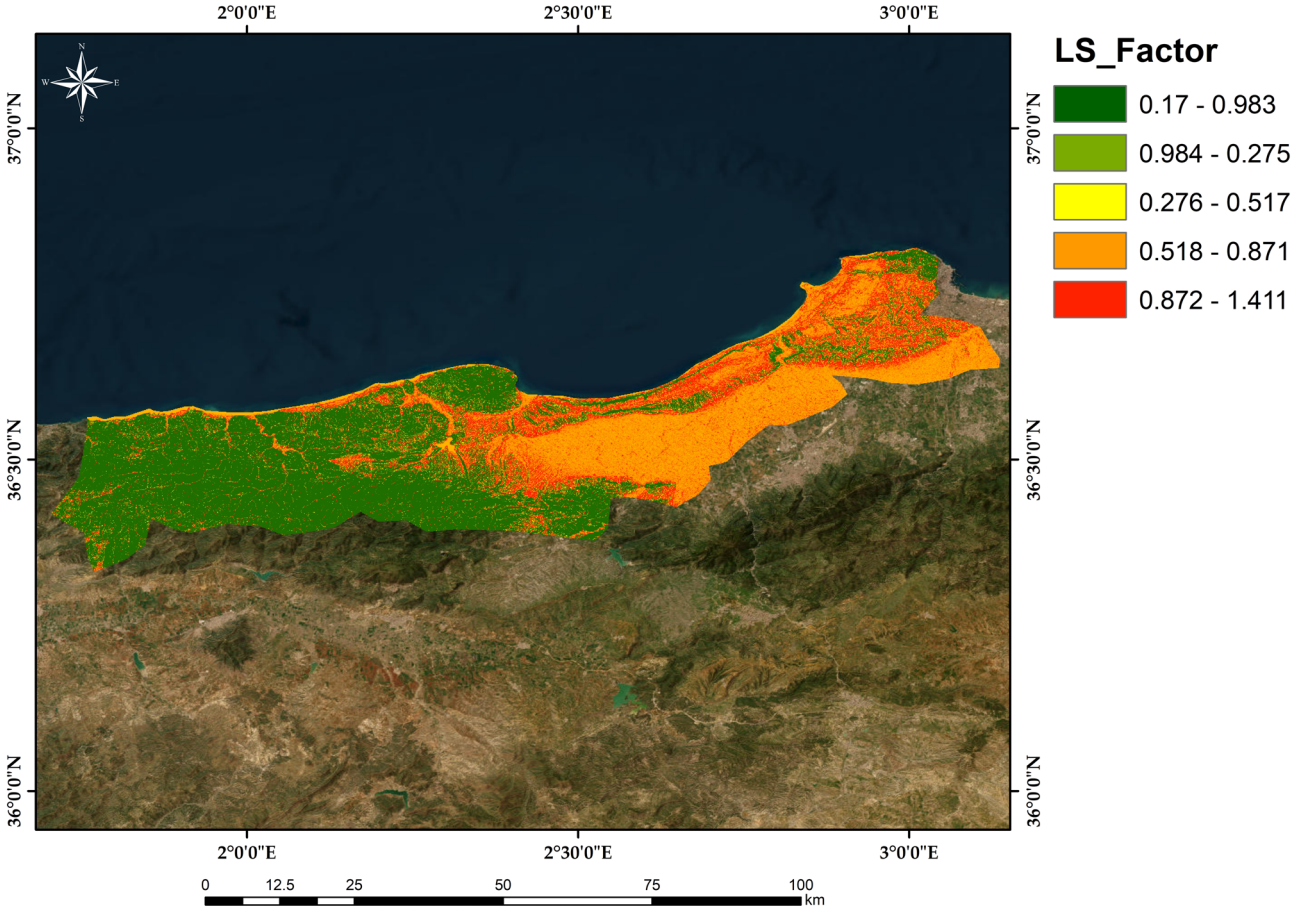

Fig. 5. The spatial distribution of topographic (L^S) factors.

region. The 86.96% accuracy rate particularly validates the weight allocation methodology through pairwise comparisons, showing that the AHP approach suggests the most influential erosion factors.

Discussion

This integrated study combining RUSLE, GIS, AHP, and CBG analysis provides a comprehensive assessment of soil erosion patterns and severity in the Mitidja plain, offering critical insights for sustainable land and water management under Mediterranean conditions. The multi-method approach effectively captured the spatial variability of erosion drivers, identifying hotspots where steep slopes, high rainfall erosivity, and poor vegetation cover converge to create severe erosion risk[57]. The CBG enhanced analysis allowed for high-resolution modeling of sediment dynamics, revealing hotspots where steep slopes, intense rainfall, and poor vegetation converge key factors exacerbating erosion in such climates[58]. The AHP-weighted integration of parameters further refined risk zonation, demonstrating that nearly 41% of the plain falls under severe erosion risk, demanding prioritized intervention, these findings underscore the importance of adaptive conservation strategies, particularly in agricultural zones where unsustainable practices amplify vulnerability[59]. By quantifying sedimentation rates and erosion susceptibility, this study provides a replicable framework for Mediterranean basins facing similar climatic and anthropogenic pressures, while emphasizing the role of cloud-based geospatial tools in scalable erosion monitoring[60].

The spatial variability of rainfall-runoff erosivity

The spatial variability of R^E factor across the Mitidja plain, ranging from 583.86 to 1072.26 MJ mm ha^{-1} h^{-1} yr^{-1} (avg. 828.06), reflects the region's complex orographic influences under Mediterranean climate conditions, and the observed west-to-east gradient, with higher erosivity in western sectors, aligns with typical precipitation patterns where mountain ranges (e.g., the Tell Atlas) create rain shadows on leeward slopes[61]. The maximum R^E values (> 1000 MJ mm ha^{-1} h^{-1} yr^{-1}) in western zones correspond to areas exposed to moisture-laden winds from the Mediterranean, consistent with study by TATLI (2025)[62] in similar Mediterranean basins. The CHIRPS-derived estimates (830–2250 mm annual rainfall) demonstrate good agreement with ground observations in northwestern Algeria[63], validating their use for RUSLE modeling. However, the lower southeastern values highlight the need for localized conservation strategies, as reduced rainfall does not necessarily equate to lower erosion risk when combined with other factors such as poor vegetation cover[64].

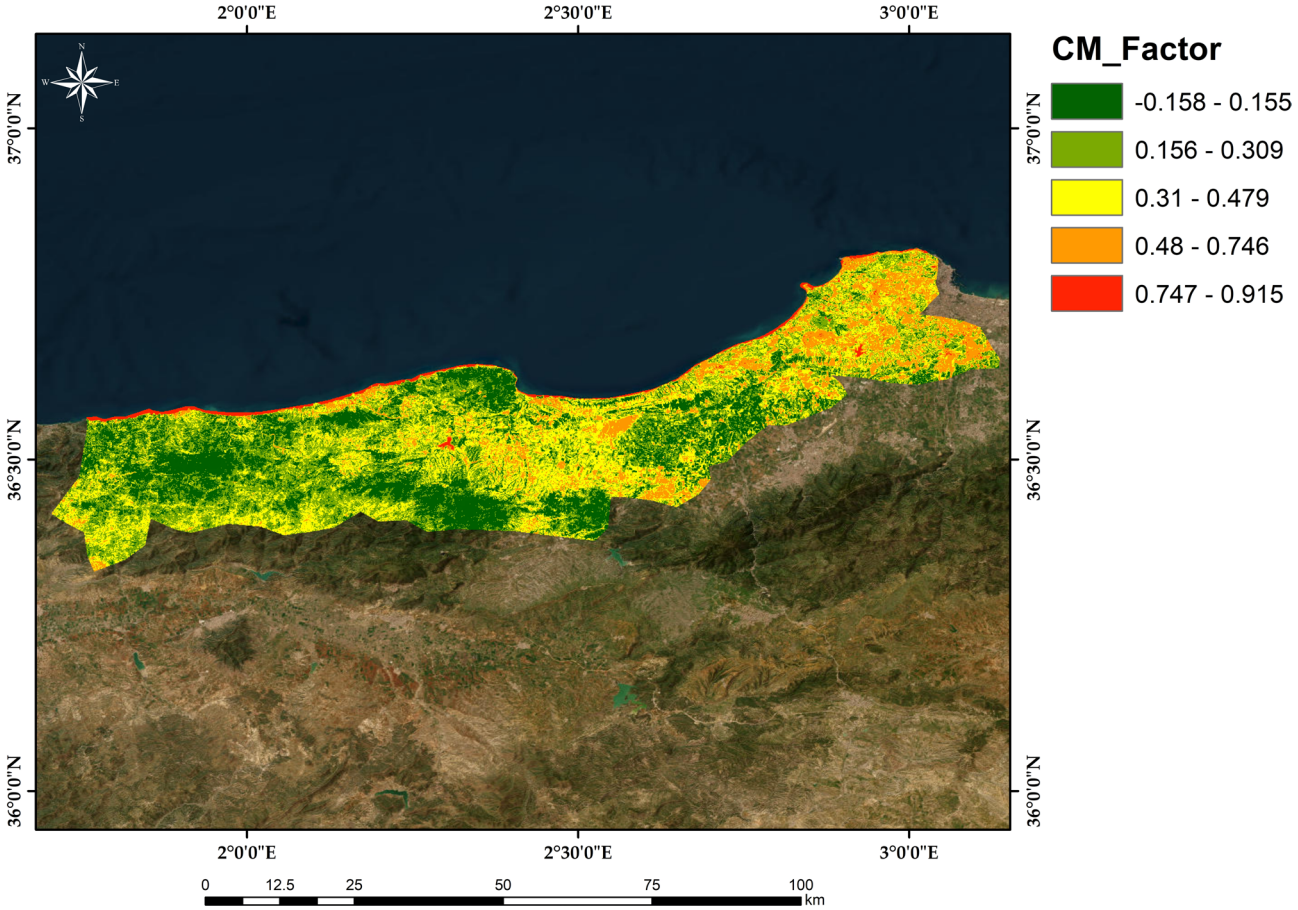

Fig. 6. The spatial distribution of cover management (C^M) factor.

The soil erodibility factor assessment

The K^S factor assessment in the Mitidja plain highlights the critical role of intrinsic soil properties particularly texture, organic matter, and structure in determining erosion susceptibility, as demonstrated by the spatial variability of K^S values (0–0.0394.0394 t ha MJ^{-1} mm^{-1}, avg. 0.0197)[65]. The higher K^S values in areas with loamy or sandy clay loam textures align with global studies showing that silt-rich soils are highly erodible due to weak cohesion and susceptibility to crusting, while clayey soils exhibit lower K^S values owing to their structural stability[33]. The integration of RUSLE with cloud-based geospatial tools (e.g., Google Earth Engine) enabled high-resolution mapping, revealing erosion hotspots where poor vegetation cover and steep slopes exacerbate soil loss, consistent with findings in similar Mediterranean basins[65]. The K^S map's utility in prioritizing conservation efforts is underscored by its correlation with field-observed erosion patterns, validating the methodology's reliability[66].

The topographic factor analysis

The L^S factor analysis revealed pronounced spatial variability (1.7–4110) across the Mitidja plain, with steeper slopes (> 22°) exhibiting significantly higher erosion potential (0.872–1.411) compared to gentle slopes (< 22°; 0–0.983.983), consistent with Mediterranean erosion studies by Othmani et al., 2023[65]. These findings emphasize the critical role of slope geometry in erosion processes, where longer, steeper slopes accelerate runoff and sediment transport[67]. The L^S map's identification of high-risk zones aligns with observed gully formations in northern mountainous areas, validating the SRTM-DEM-based approach[68]. The results underscore the necessity of terracing and contour farming in high L^S areas to mitigate slope-driven erosion.

The cover management factor analysis

The C^M factor analysis demonstrated a clear inverse relationship between vegetation density (NDVI) and erosion risk across the Mitidja plain, with values ranging from < 0.155 (dense vegetation) to 0.915 (bare soil). These results corroborate previous Mediterranean studies by García-Ruiz et al., 2013[69], that showing the vegetation cover can reduce soil loss by up to 90% compared to bare surfaces. The spatial patterns align with findings by Amiri and Pourghasemi, 2022[70], where Sentinel-2 NDVI effectively captured seasonal vegetation dynamics impacting erosion. The identification of high C^M factor zones (≥ 0.8) provides critical targets for conservation measures such as cover cropping or reforestation, particularly where steep slopes compound vegetation-deficient erosion risks[71].

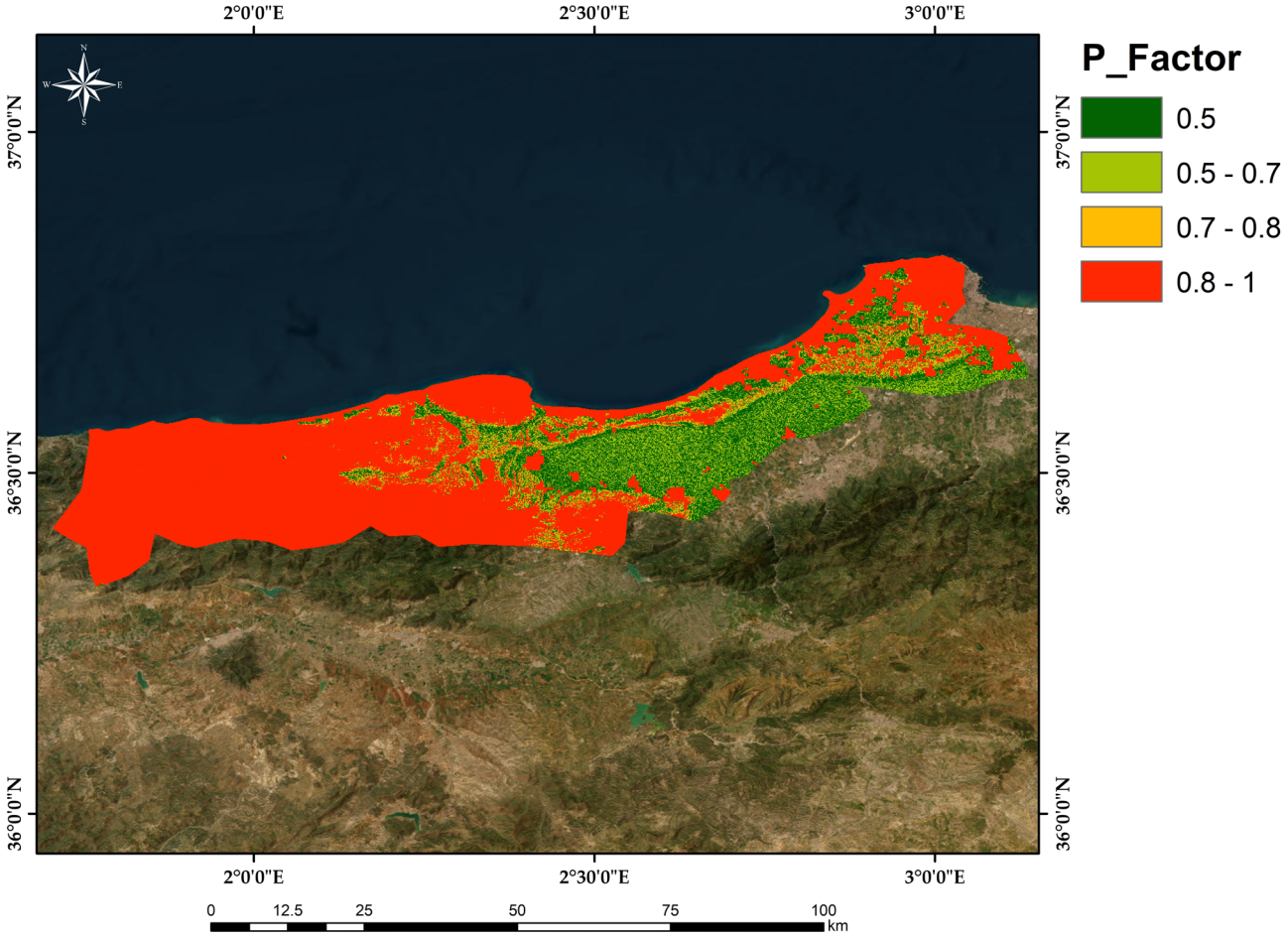

Fig. 7. The spatial distribution of conservation practice (C^P) factor.

Severity No.	Class	Average soil loss rate (t ha^{-1} yr^{-1})	Area (ha)	Area (%)
1	Moderate	38.32	942	37
2	High	65.79	337	13
3	Very high	89.63	222	9
4	Severe	161.13	1060	41

Table 9. Area statistics of soil loss zone covered by each severity class.

The conservation practice factor analysis

The C^P analysis revealed significant spatial disparities in erosion control implementation across the Mitidja plain, with unprotected mountainous and barren areas (C^P=1) showing particular vulnerability. These findings align with global studies demonstrating that human interventions can reduce soil loss by 30–80% compared to untreated areas[72,73]. The assignment of C^P =1 to natural forests, while initially counterintuitive, follows standard RUSLE protocols recognizing that passive protection differs from active conservation measures[74]. These results underscore the urgent need for terracing, check dams, and other structural measures in high-priority zones (C^P≥0.8), particularly where steep slopes compound conservation deficits[69].

The RUSLE-based soil loss estimation

The RUSLE-based soil loss estimation for the Mitidja plain indicates high erosion rates (88.72 t ha^{-1} yr^{-1} average, up to 161.13 t ha^{-1} yr^{-1}) that significantly exceed the tolerable threshold of 100–120 t ha^{-1} yr^{-1} for Mediterranean soils[75]. The spatial patterns, with 41% of the area classified as severe erosion risk, mirror findings in similar Mediterranean agroecosystems where intensive agriculture and steep slopes accelerate soil loss[75]. The cloud-based integration of RUSLE factors was used to identify critical hotspots, particularly where high rainfall erosivity (R^E factor) coincides with steep topography (L^S factor) and poor vegetation cover (C^M factor) and a dangerous combination was results by[47]. While the model's accuracy is supported by field-observed gullying in predicted severe zones, the upper range estimates (≥ 160 t ha^{-1} yr^{-1}) may reflect localized extreme events rather

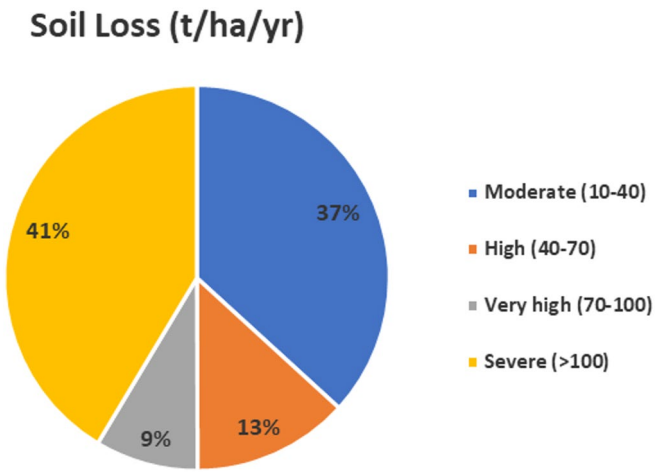

Fig. 8. Area covered by each soil loss severity class.

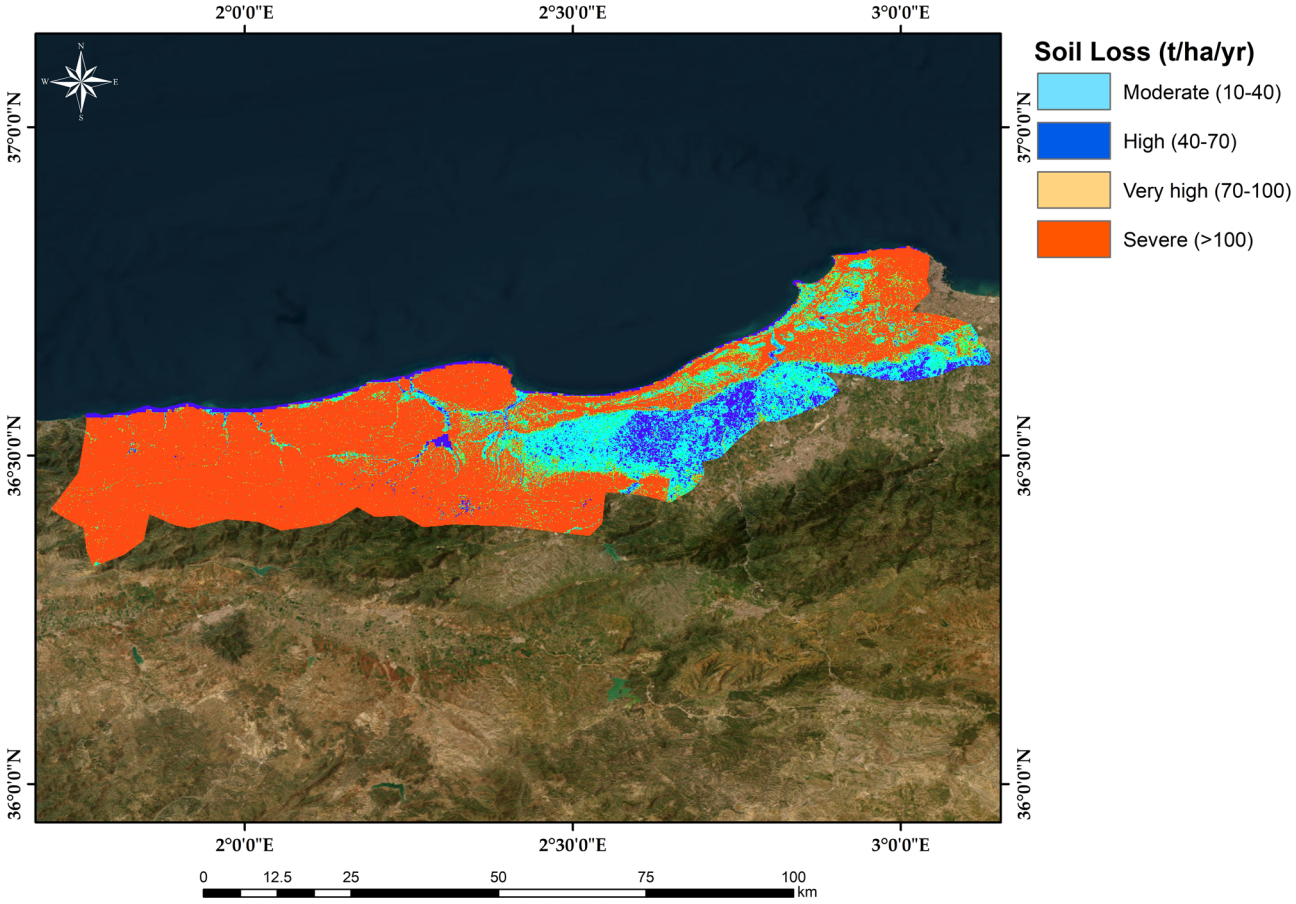

Fig. 9. The spatial distribution of of annual soil loss (RUSLE model) in the study area.

than sustained rates, as noted by Borrelli et al., 2021[17] in similar contexts. Future monitoring should incorporate seasonal vegetation dynamics to refine C^M factor estimates and improve prediction accuracy.

Soil erosion potential hazard assessment

The integrated RUSLE-AHP assessment of soil erosion hazards in the Mitidja plain effectively identified critical risk zones (22% of the area) where topographic steepness (highest AHP weight: 0.292) and intense rainfall interact with vulnerable land cover, consistent with findings in similar Mediterranean environments[24]. While forest cover reduced risk in 30% of low-vulnerability areas, the concentration of critical zones (1812 ha) along

Factor						
Parameter	R^E	K^S	L^S	C^M	C^P	Ratings
Rainfall-runoff erosivity	1	1.00	1.00	3.00	4.00	0.279
Soil erodibility	1.00	1	1.00	2.00	4.00	0.258
Topographic	1.00	1.00	1	3.00	5.00	0.292
Cover management	0.33	0.5	0.33	1	2.00	0.11
Conservation practice	0.25	0.25	0.2	0.5	1	0.061
CR	0.6					

Table 10. Pair-wise comparison matrix of different factors.

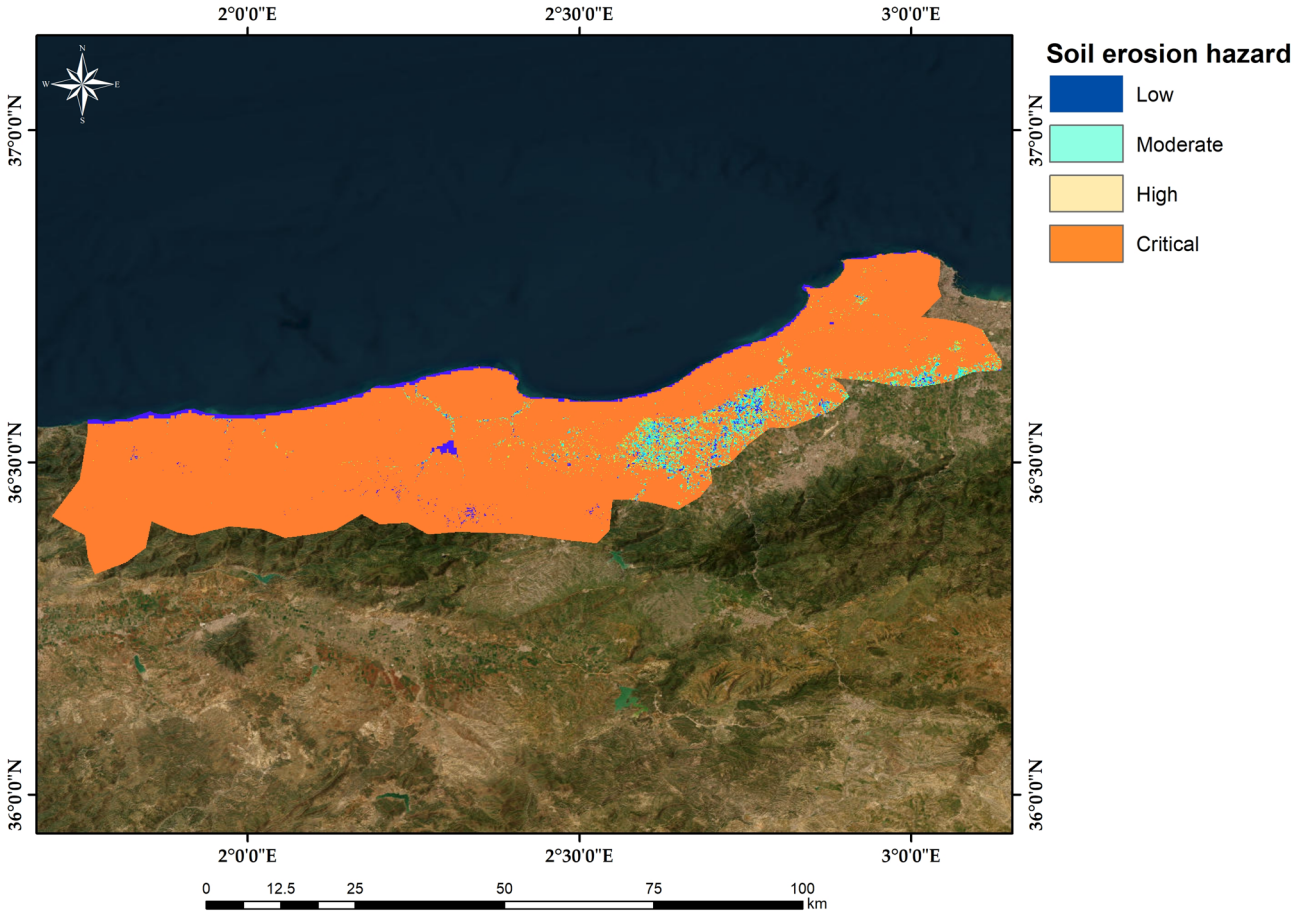

Fig. 10. The spatial patterns of soil erosion potential hazards.

Severity No.	Soil erosion hazard class	Area (ha)	Area (%)
1	Low	239	9.33
2	Moderate	250	9.76
3	High	260	10.15
4	Critical	1812	70.75
Total		2561	100

Table 11. Soil erosion hazard class.

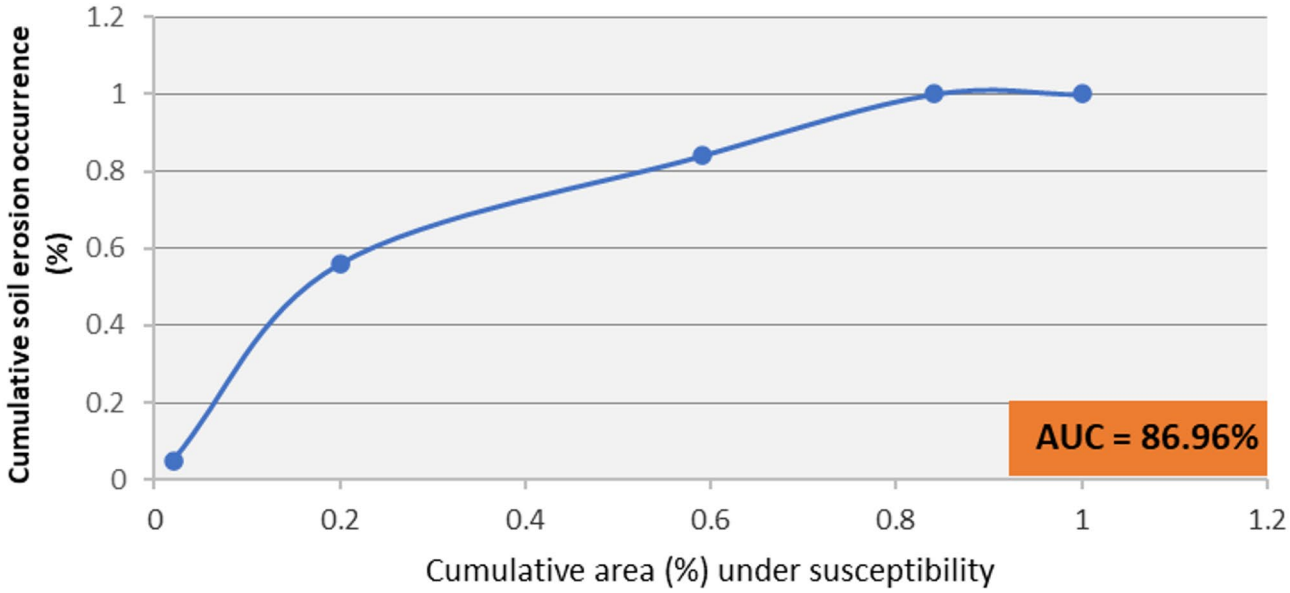

Fig. 11. Validation of soil erosion susceptibility according to AUC model.

geologically active lineaments emphasizes the need for structural interventions like check dams, as implemented in comparable settings[25]. The 48% moderate-high risk agricultural lands urgently require conservation agriculture adaptations, building on demonstrated erosion reduction strategies from Mediterranean landscapes[69].

This study advances Mediterranean erosion risk modeling by quantitatively weighting parameter contributions while highlighting the need for ongoing monitoring to address potential underestimation of seasonal vegetation effects[60].

Soil erosion model validation

The exceptional predictive accuracy of the AHP model (AUC = 0.87) confirms its robustness for soil erosion assessment in Mediterranean environments, outperforming similar studies that typically achieve AUC values of 0.75–0.85 in comparable settings[6]. This strong performance 86.96% accuracy, particularly validates the pairwise comparison weighting method, demonstrating that slope characteristics, land cover, and soil properties were optimally prioritized as dominant erosion drivers - consistent with findings by Busico et al., 2023[76] in semi-arid region. The high AUC value suggests the model effectively captured the non-linear interactions between physical factors that amplify erosion risk, as observed in other GIS-based AHP applications[77]. However, the remaining 13% uncertainty likely reflects unquantified anthropogenic factors such as tillage practices, indicating potential for improvement through incorporation of farming intensity data[78]. These results position the AHP-RUSLE integration as a reliable decision-support tool for land managers, though periodic recalibration with updated land use data is recommended to maintain accuracy[79].

Limitations and considerations

The RUSLE model, while robust, is an empirical model that may simplify complex, non-linear erosion processes. The accuracy of our estimates is inherently tied to the resolution and accuracy of the satellite-derived input data (e.g., SRTM DEM, CHIRPS precipitation). Furthermore, the C^M and C^P factors, though derived from remote sensing, represent generalized land cover classes and assumed management practices; they may not capture fine-scale, seasonal variations in vegetation cover or specific, localized farming techniques. The proposed conservation strategies (e.g., terracing, check dams) are based on model outputs and best practices but their cost-effectiveness and long-term sustainability under local socio-economic conditions remain to be evaluated through field implementation and monitoring.

Practical implications

The results provide clear and actionable insights for soil conservation in the Mitidja Plain. Severe and critical erosion zones, particularly on steep slopes and poorly vegetated agricultural lands, require urgent structural interventions such as terracing, contour farming, and check dams to reduce runoff and soil loss. Moderately vulnerable agricultural zones (48% of the plain) would benefit from reforestation, cover cropping, and conservation tillage to enhance soil protection. Unprotected mountainous zones covering 22% of the study area demand stricter land-use regulations and targeted slope stabilization measures. At the policy level, the findings support the development of evidence-based soil management frameworks that integrate erosion risk mapping into regional planning, enabling decision-makers to allocate resources efficiently and prioritize high-risk areas.

By explicitly linking erosion hotspots to practical management strategies, this study demonstrates the applied value of combining RUSLE, AHP, GIS, and cloud-based geospatial tools for soil erosion risk assessment. The approach not only enhances spatial resolution and computational efficiency but also establishes a replicable

framework for other Mediterranean regions facing similar erosion challenges. Integrating technical insights with practical recommendations ensures that the findings can inform sustainable land management, agricultural productivity, and environmental conservation across erosion-prone landscapes.

Conclusions

This study applied an integrated approach combining RUSLE, GIS, AHP, and CBG analysis within the GEE platform to assess soil erosion patterns and risks across the Mitidja plain under Mediterranean conditions. The multi-method approach revealed significant spatial variability in erosion drivers, with topographic steepness, rainfall intensity, and poor vegetation cover emerging as dominant factors contributing to severe erosion in 41% of the study area. The high-resolution mapping identified critical hotspots where converging risk factors suggest a high priority for conservation, particularly in agricultural zones and steep mountainous regions. The model validation demonstrated strong predictive accuracy (AUC = 0.87), confirming the effectiveness of the weighted parameter approach. These findings provide valuable insights for targeted land management strategies, emphasizing the need for conservation agriculture, structural measures, and vegetation restoration in high-risk areas. Overall, the research advances erosion modeling in Mediterranean environments by demonstrating the potential of cloud-based geospatial tools for efficient, large-scale analysis. The framework is replicable and adaptable to other regions with similar climatic and land-use pressures, providing valuable support for sustainable agricultural productivity, ecosystem protection, and long-term soil resource management.

Data availability

Data is provided within the manuscript.

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Author contributions

M. E. F. wrote the original draft of the manuscript, worked on conceptualization, data curation, methodology, managed the project administration, resources, software, supervision, validation, and review; M.Z. and R.L. worked on conceptualization, data curation, formal analysis; and did investigation; K.Y.F., D.M.A., W.R.Z, M.M.A.M, N.Y.R., D.E.K., A.P., A.S.E. and E.A.A. worked on conceptualization, data curation, methodology, managed the project administration, resources, software, supervision, validation, and review.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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